

Sudoku is a mathematical puzzle game. In Sudoku, we have a 9x9 grid with numbers (from 1 to 9). Additionally, this grid is divided into 9 3x3 small grids. The initial grid is filled in with some numbers, and the final solution is a grid full of numbers. See figure 1 as an example. You have the following restrictions:

- A number will appear once in each column.
- A number will appear once in each row.
- A number will appear once in each 3x3 small grid

This assignment focuses on solving Sudoku using **Backtracking**, an algorithm for Constraint Satisfaction Problems (CSPs). You will move from general state-space search to a more specialized approach where constraints guide the search.

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

5	3	4	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9	2	4	8	5	6
9	6	1	5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

Figure1: A typical sudoku puzzle seen to the left, and its solution to the right.

Objective

Your task is to implement a Sudoku solver that finds the solution for 10 different puzzles. You must treat Sudoku as a CSP where:

- **Variables:** The empty cells in the 9x9 grid.
- **Domains:** The digits 1–9.
- **Constraints:**
 - **Row Constraint:** Each number must appear once per row.

- **Column Constraint:** Each number must appear once per column.
- **Box Constraint:** Each number must appear once per 3x3 sub-grid.